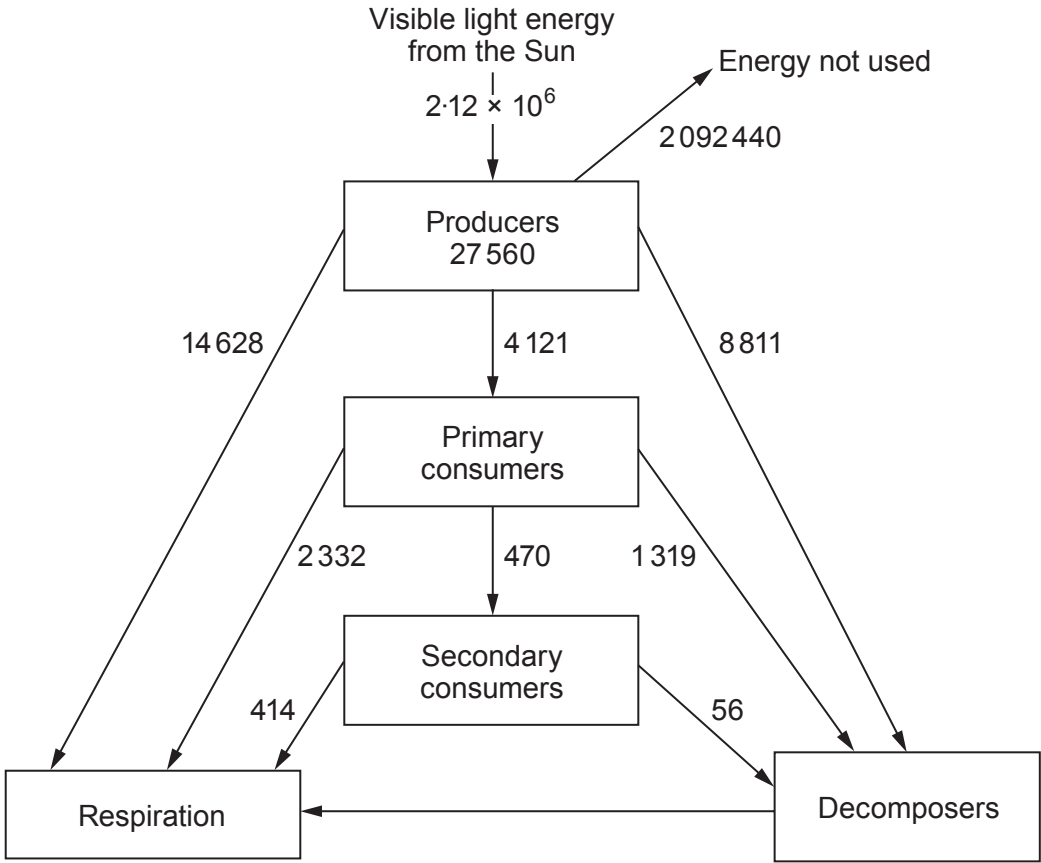


2. The mean energy flow per year through a tropical forest ecosystem in Peru, South America was measured. The results are shown in the diagram below in $\text{kJ m}^{-2} \text{ year}^{-1}$.



(a) (i) Give **one** reason why not all of the solar energy falling on the producers is used to form biomass. [1]

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(ii) Using the information given in the diagram, calculate the percentage of energy from the Sun that is fixed by photosynthesis. [2]

Percentage of energy = %



- (b) Planetary boundaries are designed to define a safe operating space for humanity to control the stability of the land, atmosphere and the sea.

One such boundary that has been crossed is the Biodiversity Boundary. This is measured in the number of species becoming extinct per million species per year. The boundary is 10 species but the current loss is over 100 species per year.

In this area of Peru, the rainforest trees are harvested using the strip felling technique. This involves harvesting the trees in an area 50m wide. That area is then left uncut for forty years allowing natural regeneration of the rainforest; this is a sustainable alternative to felling trees on a mass scale.

The photograph below shows an area of strip felling five years after harvesting.



Explain why this method of tree harvesting is helping to reduce the extinction rate whereas mass felling of trees is not. [4]

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